# Divisive Algorithms

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Divisive algorithms are inverse from agglomerative.

The $t$-th clustering contains $t+1$ clusters.

In the sequel $C_{tj}$ will denote the $j$-th cluster of the $t$-th clustering $R_t$, $t = 0, \ldots, N-1$, $j = 1, \ldots, t+1$.

Let $g(C_i, C_j)$ be a dissimilarity function defined for all possible pairs of clusters.

The initial clustering $R_0$ contains only the set $X$, that is, $C_{01} = X$.

To determine the next clustering, we consider **all possible pairs** of clusters that form a partitioning of $X$.

Among them we choose the pair, denoted by $(C_{11}, C_{12})$, that **maximizes** $g$.[1]

[1] It can be used a similarity function. In that case we should choose the pair of clusters that minimizes $g$.

4 / 17

These clusters form the next clustering $R_1$, that is, $R_1 = \{C_{11}, C_{12}\}$.

At the next time step, we consider all possible pairs of clusters produced by $C_{11}$ and we choose the one that maximizes $g$. The same procedure is repeated for $C_{12}$.

Assume now that from the two resulting pairs of clusters, the one originating from $C_{11}$ gives the larger value of $g$. Let this pair be denoted by $(C_{11}^1, C_{11}^2)$.

Then the new clustering, $R_2$ consists of $C_{11}^1$, $C_{11}^2$, and $C_{12}$.

Relabeling these clusters as $C_{21}, C_{22}, C_{23}$, respectively, we have

$$R_2 = \{C_{21}, C_{22}, C_{23}\}$$

Carrying on in the same way, we form all subsequent clustering.

- Initialization
  - Choose $R_0 = \{X\}$ as the initial clustering.
  - $t = 0$
- Repeat
  - $t = t + 1$
  - For $i = 1$ to $t$
    - Among all possible pairs of clusters $(C_r, C_s)$ that form a partition of $C_{t-1,i}$, find the pair $(C_{t-1,i}^1, C_{t-1,i}^2)$ that gives the maximum value for $g$.
  - Next $i$
  - From the $t$ pairs defined in the previous step choose the one that maximizes $g$. Suppose that this is $(C_{t-1,j}^1, C_{t-1,j}^2)$.
  - The new clustering is

$$R_t = (R_{t-1} - \{C_{t-1,j}\}) \cup (C_{t-1,j}^1, (C_{t-1,j}^2)$$

  - Relabel the clusters of $R_t$.
- Until each vector lies in a single distinct cluster.

6 / 17

Different choices of $g$ give rise to different algorithms.

One can easily observe that this divisive scheme is computationally very demanding, even for moderate values of $N$. This is its **main drawback**, compared with the agglomerative scheme.

Thus, if these schemes are to be of any use in practice, some further computational simplifications are required:

- One possibility is to make compromises and not search for all possible partitions of a cluster.

## Split criterion

Let $C_i$ be an already formed cluster. Our goal is to **split** it further, so that the two resulting clusters $C_i^1$ and $C_i^2$, are as **dissimilar** as possible.

Initially, we start with

$$C_i^1 = \emptyset \qquad \wedge \qquad C_i^2 = C_i$$

Then, we identify the vector in $C_i^2$ whose **average dissimilarity** from the remaining vectors is maximum, and we move it to $C_i^1$.

In the sequel, for each of the remaining $x \in C_i^2$, we compute its average dissimilarity with the vectors of $C_i^1$:

$$g(x, C_i^1)$$

as well as its average dissimilarity with the rest of the vectors in $C_i^2$:

$$g(x, C_i^2 - \{x\})$$

8 / 17

## Split criterion

If for every $x \in C_i^2$, it holds

$$g(x, C_i^2 - \{x\}) < g(x, C_i^1)$$

then we stop.

Otherwise, we select the vector $x \in C_i^2$ for which the difference

$$D(x) = g(x, C_i^2 - \{x\})) - g(x, C_i^1)$$

is **maximum** (among the vectors of $C_i^2$, $x$ exhibits the maximum dissimilarity with $C_i^2 - \{x\}$ and the maximum similarity to $C_i^1$) and we move it to $C_i^1$.

The procedure is repeated until the termination criterion is met.

One important issue in hierarchical algorithms is determining the best clustering within a given hierarchy. This is equivalent to identifying the **number of clusters** that **best fits** the data.